

Detection of Acute Lymphoblastic Leukemia Using a Novel Bone Marrow Image Segmentation

M. Anline Rejula*, Ben M Jebin, Ravi Selvakumar, S. Amutha, and Eberlein George

Abstract: In our study, we present a novel method for automating the segmentation and classification of bone marrow images to distinguish between normal and Acute Lymphoblastic Leukaemia (ALL). Built upon existing segmentation techniques, our approach enhances the dual threshold segmentation process, optimizing the isolation of nucleus and cytoplasm components. This is achieved by adapting threshold values based on image characteristics, resulting in superior segmentation outcomes compared to previous methods. To address challenges, such as noise and incomplete white blood cells, we employ mathematical morphology and median filtering techniques. These methods effectively denoise the images and remove incomplete cells, leading to cleaner and more precise segmentation. Additionally, we propose a unique feature extraction method using a hybrid discrete wavelet transform, capturing both spatial and frequency information. This allows for the extraction of highly discriminative features from segmented images, enhancing the reliability of classification. For classification purposes, we utilize an improved Adaptive Neuro-Fuzzy Inference System (ANFIS) that leverages the extracted features. Our enhanced classification algorithm surpasses traditional methods, ensuring accurate identification of acute lymphoblastic leukaemia. Our innovation lies in the comprehensive integration of segmentation techniques, advanced denoising methods, novel feature extraction, and improved classification. Through extensive evaluation on bone marrow samples from the Acute Lymphoblastic Leukemia Image DataBase (ALL-IDB) for Image Processing database using MATLAB 10.0, our method demonstrates outstanding classification accuracy. The segmentation accuracy for various cell types, including Band cells (96%), Metamyelocyte (99%), Myeloblast (96%), N. myelocyte (97%), N. promyelocyte (97%), and Neutrophil cells (98%), further underscores the potential of our approach as a high-quality tool for ALL diagnosis.

Key words: Adaptive Neuro-Fuzzy Inference System (ANFIS); blood cells; segmentation; leukemia classification

1 Introduction

Many disorders are discovered using microscopic images of blood cells. Changes in a person's blood

state indicate the onset of illnesses. Leukemia can cause death if neglected or not treated. Leukemia is the sixth major cause of death for women and the fifth

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leading cause of death for men. The bone marrow is where leukemia begins. It depicts the interior of each bone, which includes a thin substance known as bone marrow^[1].

As shown in Fig. 1, the blood marrow is made up of platelets, white blood cells (leucocytes), and red blood cells (erythrocytes). White blood cells can be used to identify leukemia and help identify or fight any disease or infection. Oxygen is transported by erythrocytes from the lungs to the organs and other bodily components. The elements that support the clotting process are platelets, which in turn stop the excess bleeding.

Leukemia, or infected cells, is the outcome of bone marrow cells changing and becoming infected. Several characteristics set these leukemia cells apart from healthy ones. Their development pattern is aberrant, and they live longer than regular cells. Normal cells die after a given length of time, whereas leukemia cells do not^[2]. While new leukemia cells are produced improperly, old leukemia cells exist for a very long time. The rate at which leukemia cells advance varies depending on the kind of leukemia.

Figure 2 depicts how a stem cell can develop into a myeloid or lymphoid stem cell. A myeloid stem cell gives rise to a myeloid blast. The myeloid blast produces red blood cells (erythrocytes), white blood cells (leucocytes), and platelets. In the same way, lymphoid system cells also produce white blood cells by evolving into lymphoid blasts. The five different categories of White Blood Cells (WBC) that can be found in the blood are lymphocytes, myelocytes, neutrophils, basophils, and eosinophils. For leukemia

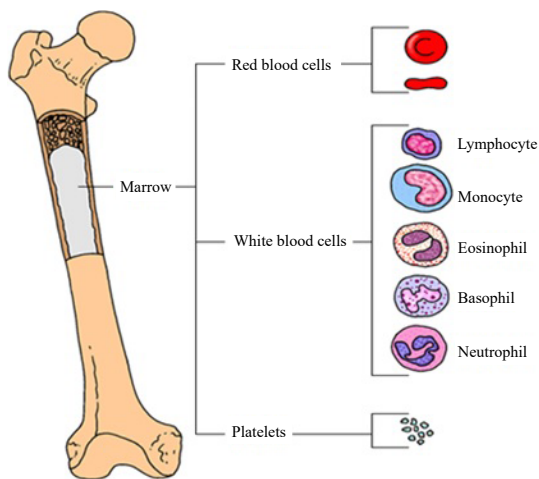


Fig. 1 Bone marrow.

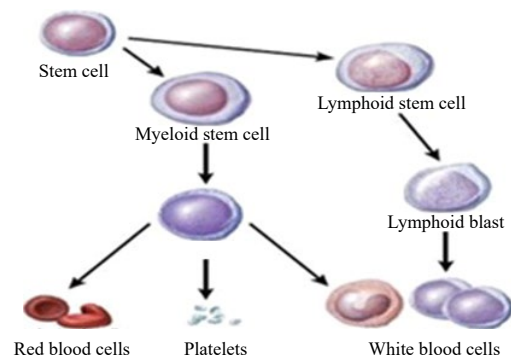


Fig. 2 Stem cell evolution.

patients, the bone marrow creates abnormal WBCs, where their life span is increased; they multiply abnormally and interrupt the functions of the normal cells.

Detecting leukemia at an early stage is the most effective technique to save the patient's life. However, hand counting is done in the conventional way, which is tedious and time-consuming. The Automated Hematology Counter is then created, although it is quite costly and unaffordable for developing and underdeveloped nations^[1]. Enhanced thresholding has advantages over standard segmentation techniques like watershed or region growth, but it relies on the application and the properties of the pictures being segmented.

Traditional segmentation techniques can handle pictures with complicated structures and a broad range of intensity fluctuations, making them more robust and adaptable. For instance, region expanding may be used to divide items with irregular geometries, whereas watershed segmentation can be used to segment objects that are touching or overlapping. Enhanced thresholding, on the other hand, is a less complicated and quicker technique that is effective for pictures with well-defined intensity thresholds, such as binary or grayscale images with distinct object borders. Additionally, it may be employed when the item of interest has a uniform intensity distribution, which makes it simple to recognize the object from its intensity values. Enhanced thresholding, however, may not be appropriate for more complicated pictures since it might cause over- or under-segmentation if the intensity values are poorly defined or if the item of interest has a large range of intensities. This may cause segmentation problems and compromise the precision of subsequent analyses. On order to identify defective WBCs in leukemia slides, we have developed an

automated approach. For the identification and classification of leukemia cells, we employed (1) enhanced dual threshold segmentation for leukocyte cell image segmentation; (2) hybrid DWT feature extraction used for leukocyte cell image; and (3) an improved Adaptive Neuro-Fuzzy Inference System (ANFIS) classifier. As a result, computing time is cut down, and efficiency and accuracy are improved^[3].

Our contributions can be summarized as follows:

(1) Automated approach: The authors have developed an automated method for identifying defective WBCs in leukemia slides. This is an important contribution as it improves the efficiency and accuracy of diagnosing leukemia, which is traditionally a time-consuming and subjective process^[4].

(2) Enhanced dual threshold segmentation: The authors have introduced an improved technique for segmenting leukocyte cell images called enhanced dual threshold segmentation. This method goes beyond traditional thresholding techniques by considering both the nucleus and cytoplasm components of the blood smear images. This more accurate segmentation enables better analysis and classification of WBCs.

(3) Hybrid Discrete Wavelet Transform (DWT) feature extraction: The authors have employed a hybrid discrete wavelet transform to extract features from leukocyte cell images. This approach combines wavelet analysis and feature extraction, allowing for more comprehensive and informative characterization of the cells. The use of hybrid DWT enhances the ability to capture relevant features for classification purposes.

(4) Improved Adaptive Neuro-Fuzzy Inference System (I-ANFIS) classifier: The authors have utilized an enhanced adaptive neuro-fuzzy inference system as a classifier in their automated approach. This classifier combines fuzzy logic principles with neural network techniques to effectively classify normal and abnormal samples. The improvements made to the ANFIS system enhance its accuracy and robustness in distinguishing between different types of WBCs.

Overall, the novelty of this work lies in the combination of enhanced dual threshold segmentation, hybrid DWT feature extraction, and improved ANFIS classification for automated leukaemia cell analysis. These innovative elements contribute to more accurate and efficient leukaemia diagnosis, offering potential advancements in medical image analysis and disease detection.

2 Related Work

Madhlloom et al.^[5] used image arithmetic and threshold procedures to detect the white blood cell nuclei. The difficult part of building this system is deciding on a suitable segmentation threshold. The nuclei of white blood cells cannot be effectively segmented using the threshold used in the established approach. The focus of this study is on segmenting the nucleus of white blood cells, as the nucleus features are usually sufficient for cell type identification. The proposed method combines automatic contrast stretching, image arithmetic operations, minimum filtering, and global thresholding techniques to separate the nucleus from the entire cell body. The results demonstrate promising accuracy levels ranging from 85% to 98% when compared to expert analysis. However, it is important to acknowledge some limitations of the proposed method. One disadvantage is that the accuracy of the segmentation results may vary depending on the image quality and characteristics of the blood samples. Factors, such as staining variations, overlapping cells, and image artifacts, can affect the segmentation performance. Additionally, the proposed method might encounter challenges when dealing with abnormal or rare cell types that require more specialized analysis techniques.

From a blood image, Kovalev et al.^[6] devised a technique to categorise five categories of leucocytes. They started by detecting nuclei and then used region-growing methods to uncover the whole membrane. The results of the evaluation demonstrate high levels of accuracy in different aspects of recognition. The method achieves recognition accuracy of no less than 98% when tested on a dataset consisting of 662 images representing five types of WBCs. These images are obtained from 22 different blood samples using various imaging systems. Conversely, it is important to consider some potential disadvantages of the proposed method. One limitation could be the reliance on specific imaging systems and sample sources. The effectiveness of the method may vary when applied to images obtained from different imaging systems or with variations in sample preparation methods. Another potential drawback is the limited scope of the study, as it focuses on differentiating only five types of WBCs. While achieving high accuracy for these specific cell types is valuable, the method may not be equally effective for recognizing rarer or abnormal cell variants.

Scotti et al.^[7] used a low-pass filter to eliminate background noise, threshold procedures, and clustering to separate white blood cells were all used. He attempted to produce strong segmentation results for photographs captured in a variety of lighting settings. The authors focus on enhancing microscope images by effectively removing unwanted background elements. They also proposed a novel segmentation strategy that reliably identifies white blood cells, enabling the extraction of relevant features for automated disease diagnosis. By implementing these techniques, the paper showcases the potential to improve the accuracy of leukaemia identification by leveraging the extracted features from the segmented white blood cells. This approach provides a valuable contribution to the field of automated disease diagnosis, particularly in the context of leukaemia. One potential drawback is the reliance on specific microscope imaging setups or techniques. The effectiveness of the proposed methods may vary when applied to images obtained from different microscope models or with variations in sample preparation protocols.

White blood cell segmentation was accomplished by Piuri and Scotti^[8]. They employed an edge detection approach for each leucocyte, used a neural network for categorisation, and trained a neural network to differentiate lymphoblasts using morphological clues. Moreover, training a neural network solely based on morphological clues for differentiating lymphoblasts may have limitations. Morphological features alone may not provide sufficient discriminatory information to accurately classify lymphoblasts from other cell types. Other complementary features, such as cytoplasmic staining patterns or genetic markers, may be required to improve the classification accuracy and robustness.

In a microscopic blood image, Halim et al.^[9] suggested an automated approach for counting the number of blasts. The findings are pretty impressive. However, the system's flaw is that no mechanism for determining the best threshold for better segmentation is stated. No feature is retrieved, and no classifier is applied. Another problem with the suggested system is that it does not have a way to collect features or a classifier. Feature extraction is the process of getting useful information or traits from the divided items, which can help with classifying them or doing more research on them. By adding things like form, colour, or strength, it is possible to get a more complete picture

of the blast cells. Also, a classifier can be used to tell blast cells apart from other types of cells or to divide blast cells into groups.

Mohapatra et al.^[10] employed clustering to segment white blood cells and obtained traits, such form, colour, texture, fractal, Fourier descriptors, and contour. The software is designed to find leukaemia. The results are reasonable, but because they come from a private dataset, they cannot be compared to those from other methods. The ability to assess the efficacy and efficiency of the clustering-based segmentation and the derived features for leukaemia diagnosis is further constrained by the absence of comparison with other approaches. It is difficult to judge the effectiveness or competitiveness of the suggested strategy compared to other strategies without comparison findings.

Labati et al.^[11] offered a dataset that includes normal and leukaemia patients' blood samples, which is proven to be useful for the suggested system of testing. This data collection is open to the public for research purposes.

A technique was created by Liao and Deng^[12] to extract individual white blood cells from an image. Before going on to contour identification, they employed a few threshold techniques. They made the assumption that all cells are circular in order to make the software run smoothly. Their method has an issue in that it makes the assumption that all cells are circular, which may not be accurate in practical applications. When cells have irregular forms or morphological differences, this assumption may reduce the precision of cell identification and segmentation.

Angulo and Flandrin^[13] suggested a "two-stage blood picture segmentation technique" in their system. They employed binary filtering as well as certain automated thresholding approaches. Using this method, the nucleus, cytoplasm, and nucleolus may all be retrieved from pictures of lymphocytes. Some forms of blood stains or fluctuations in picture quality may not be acceptable for binary filtering and automated thresholding techniques. Variations in the intensity and distribution of cell components brought up by various stains and imaging settings might provide subpar segmentation outcomes.

Sinha and Ramakrishnam^[14] suggested a method for automatically segmenting leucocytes. They employed Gaussian mixture modelling and the EM method. Parameter tweaking is not necessary in this procedure. Gaussian mixture modelling and EM provide a non-

supervised leucocyte segmentation method, although it may not work for all staining methods. Different staining techniques might change cell morphology and intensity distributions, requiring customised parameterization or changes for proper segmentation.

Treera-Umporn^[15] created a technique for dividing white blood cells' nuclei into distinct sections. He applied a fuzzy clustering approach. This method is effective for nucleus segmentation. However, it does not take into consideration cytoplasm extraction, which is equally as important as nucleus segmentation. The dataset may not capture real-world blood sample variety and complexity. The dataset's shortcomings may restrict the system's generalizability to various settings or populations.

Dorini et al.^[16] suggested a technique for nucleus extraction that is based on the image forest transform and the watershed transform. However, the usefulness of this method is restricted to cells with a certain shape since it is less, likely to be successful in cells, in which the cytoplasm is not spherical.

A technique for the segmentation of cells using the LUV colour space was given by Comaniciu and Meer^[17]. The extraction of cytoplasm, which is an essential part of thorough cell segmentation and analysis, is not addressed, which is a downside of this method. A Guiding Vector Field (GVF)-based approach was proposed by Yang et al.^[18] for the segmentation of leucocytes; however, the authors encountered difficulties in precisely differentiating textures and weak edges, which results in an insufficient separation of lymphocytes. Fang et al.^[19] suggested that Particle Swarm Optimisation (PSO) might be used to train a neural network for the classification of white blood cells. The disadvantage of this technology is that it could not work very well in recognising cytoplasmic nuclei, which reduces the efficiency with which it can carry out full cell segmentation^[20]. A hybrid technique that combines threshold segmentation, mathematical morphology, and fuzzy cellular neural networks was suggested by S. Wang and M. Wang^[21]. On the other hand, it does not have the capacity to successfully separate the cytoplasm and the nucleus, which is a skill that is necessary for proper cell segmentation. Chinwaraphat et al.^[22] presented an approach called modified fuzzy c-means clustering as a way of addressing doubts about the belongingness of individual cells. However, human cropping is still required for the test photos, which

indicates that the performance of the suggested approach for automatically segmenting photographs has limits.

An automated segmentation strategy that is based on pixel categorization was presented by Meurie et al.^[23], although it is possible that the average segmentation performance is not sufficiently high. Additionally, the use of many classifiers results in an increase in the typical running time, which may have an effect on the method's capability of being applied in real time.

In inference, despite the fact that the aforementioned techniques provide a number of different approaches to white blood cell segmentation, they all share a number of drawbacks, including the inability to handle a wide variety of cell shapes, variations in staining, insufficient cytoplasm extraction, and poor performance in terms of accuracy, generalizability, or computational efficiency. It is required to do more study and make changes in order to overcome these constraints, which will in turn increase the reliability and application of the suggested segmentation approaches.

3 Proposed Work

The suggested work is divided into many steps, including pre-processing, segmentation, feature extraction, and classification. A median filter is used to pre-process the blood image. Enhanced dual-threshold is used to split the pre-processed image into two parts. Using the hybrid discrete wavelet transform, features are retrieved. Additionally, classification is carried out using an improved AFIS that can distinguish between normal and leukemic cells as shown in Fig. 3.

3.1 Segmentation

The improved dual-threshold segmentation technique consists of three phases: pre-processing, threshold segmentation, and post-processing, as shown in Fig. 4, where I1 is the blood group, G2, G2', T1, and T2 are phases of Motosis, H, H2, and H2' are the haptoglobin phenotypes. Two images are acquired for further handling during the pre-processing stage: one is

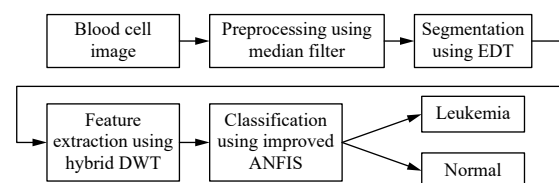


Fig. 3 Proposed work.

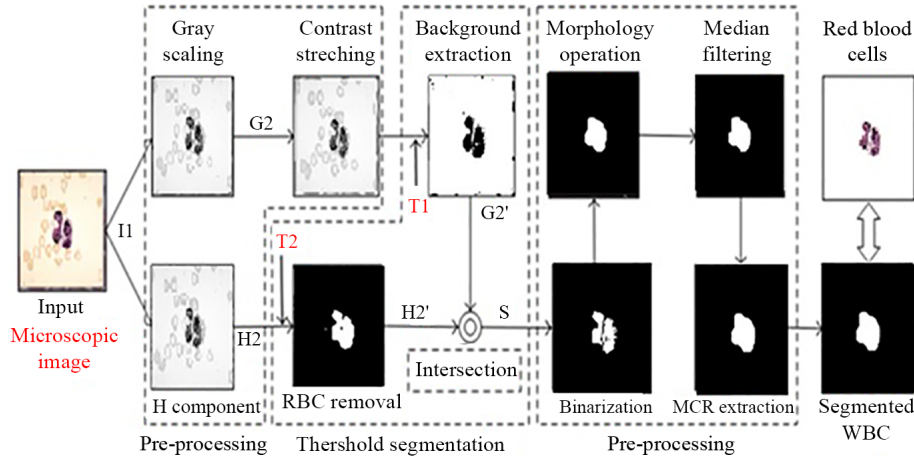


Fig. 4 Block diagram of the enhanced dual-threshold segmentation.

a contrast-stretched grey image, and the other is an H-part image from the modified YCbCr colour space^[24].

The background separation, red blood cell extraction, and the finest threshold collection are the three key steps comprised by the threshold segmentation. The best threshold collection component uses the golden section search approach. Finally, in the post-processing stage, median filtering, and arithmetical morphology are utilised to remove noise and incomplete WBCs.

3.2 Image pre-processing

Two processed images are acquired during pre-processing for subsequent image segmentation processing. Figure 5b shows how the colour input image is converted to a grayscale image. Each member of the matrix^[25] determines the intensity of each pixel. For convenience, most advanced digital files employ integers between 0 and 255, yielding 256 different grey levels. This number of grey levels is sufficient for working with medical images.

The Image Contrast Stretching (ICS) is carried out by sliding a window over the image (known as the KERNEL) and altering the inner component using Eq. (1) below:

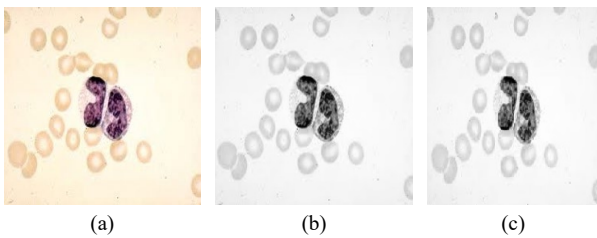


Fig. 5 (a) Original image, (b) grayscale image, and (c) image contrast stretching.

$$I_p(1+x) = \frac{255[I_o(x,y) - m_i]}{(m_a - m_i)} \quad (1)$$

where p is the pixel organize of the picture; I_p is the differentiation-extending process's yield shading level for the pixel; The lowest and highest shading levels in the information picture are m_i and m_a , respectively. I_o is the pixel's information shading level. $I_o(x,y)$ is the intensity of original image co-ordinates (x,y) in pixel.

3.3 Threshold segmentation

The system's core consists of image thresholding applications for image segmentation. Thresholding, which is applied to a single picture, is the most basic method of segmenting an image. Thresholding^[26] is a technique for creating binary pictures from a greyscale image. To isolate these modes from the background, set a T threshold that separates them. $f(x,y)$ is the intensity or pixel value of the image at the point specified by the coordinates (x,y) . Any image point (x,y) with $f(x,y) > T$ is an object point; otherwise, the point is a background point. Equation (2) defines the threshold image $g(x,y)$,

$$g(x,y) = \begin{cases} A1, & \text{if } f(x,y) > T; \\ B1, & \text{if } f(x,y) \leq T \end{cases} \quad (2)$$

where pixel A1 denotes the items, whereas pixel B1 denotes the backdrop. Through conversion, $A1 = 1$ is assigned to the white color and $B1 = 0$ is assigned to the black color in the picture. The global threshold is regarded in the equation after the complete picture, T is considered an applicable constant. Variable thresholding refers to variations in T over the course of an image.

3.4 Dual thresholding

Two image characteristics are combined using dual

thresholding. This technique divides the image based on histogram equalization^[27]. $G(x, y)$ provides a split image as follows:

$$G(x, y) = \begin{cases} A, & \text{if } f(x, y) > T_2; \\ B, & \text{if } T_1 < f(x, y) \leq T_2; \\ C, & \text{if } f(x, y) \leq T_1 \end{cases} \quad (3)$$

The split image (G) is formed by assigning different values (A, B, C) to pixels according to their intensity levels and the specified thresholds.

To come up with an acceptable objective function, start with the Dice Similarity Coefficient (DSC), which is commonly employed to quantify segmentation impact. It is described as

$$\text{DSC}(C_1, D_1) = \frac{2(C_1 \cap D_1)}{(C_1 \cup D_1)} \quad (4)$$

where C_1 is the zone of the objective district of the aftereffect of a subsequently divided image, and D_1 is the zone of the objective district of the ground truth image actuated by the physically portioned pictures. DSC varies by a factor between 0 and 1. According to this reality, the goal of work is determined by

$$\text{DSC} = f(\text{Threshold}_1, \text{Threshold}_2) \quad (5)$$

The issue of advancement is illustrated as follows: By selecting appropriate Threshold_1 and Threshold_2 , the influence DSC will be as widespread as possible.

3.5 Post processing

White platelets can be divided after edge division. In the interim, binarization is used to transform the output image into a paired image, because it is still a grayscale image with some noise^[28]. Then, morphological disintegration is utilised to fill tiny apertures, interface restricted holes, and expel small dabs, as well as the inadequate WBCs from the picture, in conjunction with middle separation and Maximum Connected Region (MCR) extraction procedures. The parallel image is next transformed into an RGB shaded image, but this step is optional.

3.6 Binarization

Binary images are frequently encountered in digital image processing tasks, often arising as masks or outcomes of various techniques like segmentation, thresholding, and dithering. Due to their straightforward nature, they find extensive use across different applications^[29],

$$f(x, y) = \begin{cases} 1, & \text{if } f(x, y) > T; \\ 0, & \text{if } f(x, y) < T \end{cases} \quad (6)$$

Depending on the significance of the threshold T in the histogram, the image intensity may be divided into two distinct groups. Background pixels $f(x, y) < T$ in the segmented binary picture have a value of zero, whereas object pixels $f(x, y) > T$ have a value of one.

3.7 Morphological operation

Particularly in machine vision and autonomous object recognition, morphological processes are crucial to advanced digital image processing^[29]. The Structuring Element (SE), controlled by a grid of 0s and 1s, determines the degree of thickening. The following Eq. (7) demonstrates how to describe dilation mathematically using set notation:

$$A \oplus A_s = \{z | (A'_s) z \cap A \neq \Phi\} \quad (7)$$

whereas A is the original image, A_s is the structuring element, A'_s represents the dilation of set A by scale factor s , and z is the resulting set and is an empty component. The action of dilation is commutative and well-known. By using an organizing component, Erosion contracts or decreases the objects in a double image. The numerical depiction of erosion is shown in the following:

$$A \oplus A_s = \{z | (A'_s) z \cap A_c \neq \Phi\} \quad (8)$$

whereas A_c is the complement of original image A . The command `imerode` is used to execute erosion in MATLAB (image name, SE). Erosion and dilation are often utilized in image processing and in different mixes. Morphological picture opening and closing are triggered by the collection of these two criteria. Morphological opening is represented as a disintegration activity followed by an expansion activity^[26].

In Eqs. (9) and (10), the terms “ $X \oslash Y$ ” and “ Y ” stand for the morphological opening of the picture formed by X/Y , where the morphological opening is defined as the erosion of X by Y followed by dilatation of the resulting result,

$$X \oslash Y = (X \ominus Y) \oplus Y \quad (9)$$

$$X \bullet Y = (X \oplus Y) \ominus Y \quad (10)$$

where $X \ominus Y$ represents the erosion of sets X and Y , $X \oplus Y$ represents the dilation of set X by set Y . $\oplus$ denotes dilation and $\ominus$ denotes erosion^[29]. Equation (11) demonstrates how an image f will first enlarge

before eroding as a result of a structural component s ,

$$f \oslash s = (f \ominus s) \oplus s \quad (11)$$

An opening is an idempotent job in which further openings with the same organizing component have no further influence on the picture after it has been opened,

$$(f \oslash s) \oslash s = f \oslash s \quad (12)$$

In Eq. (13), the dilatation followed by erosion of an image f by a structuring element S_{rot} is represented,

$$f \cdot s = (f \oplus S_{\text{rot}}) \ominus S_{\text{rot}} \quad (13)$$

3.8 Feature extraction

The following feature extraction process uses the segmented information, as well as the spatial information of the segmented objects. The 2D-Haar wavelet transformation is used to extract features from the de-noised picture for decomposition, which extracts and maintains the low-level frequency components in the LL-sub band of the decomposed image^[30].

3.9 Algorithm of Hybrid DWT for color features extraction

Figureing a colour feature is a group of bins that depict the distribution of colours in an image, with each bin corresponding to a colour in the quantized colour space^[30]. In Eq. (14), a colour histogram for a particular image is represented as a vector,

$$H = \{H[0], H[1], H[2], H[3], \dots, H[I], \dots, H[n]\} \quad (14)$$

The number of pixels in the image that contain colour I is represented by $H[i]$, the number of bins in the colour histogram is represented by n , and I is the colour bin in the colour histogram which is the i -th element of pixel. Typically, a colour histogram bin is assigned to each pixel in an image^[31]. The normalized colour histogram is shown in the following:

$$H' = \{H'[0], H'[1], H'[2], H'[3], \dots, H'[i], \dots, H'[n]\} \quad (15)$$

where $H'[i] = H[i]/p$, and p is the total number of pixels of an image.

3.9.1 Contrast

It is the measure of contrast between the intensity of a pixel and those of its neighbours throughout the whole ROI,

$$\text{Contrast} = \sum_i \sum_j (i-j)^2 C(i, j) \quad (16)$$

where $C(i, j)$ refers to the co-occurrence matrix, and the nested summations iterate over the indices i and j , i is the pixel's i -th element, and j is the pixel's j -th element.

3.9.2 Homogeneity

Homogeneity is defined as the difference between the distribution of elements in the Gray-Level Co-occurrence Matrix (GLCM) and the GLCM of each ROI, with a value ranging from 0 to 1. It is referred to as

$$\text{Homogeneity} = \sum_i \sum_j \frac{C(i, j)}{1 + |i - j|} \quad (17)$$

3.9.3 Mean

Mean is the arithmetic average of a set of values, and it is defined as

$$\text{Mean } (\mu) = \frac{\sum_i \sum_j C(i, j)}{M \times N'} \quad (18)$$

where μ is the mean. M is the number of rows, and N' is the number of columns in the array.

3.9.4 Energy

The measure of the amount of pixel pair repeats is called energy (E). It assesses an image's consistency. The energy value is high when pixels are comparable,

$$\text{Energy } (E) = \sum_i \sum_j C(i, j)^2 \quad (19)$$

3.9.5 Area

In Eq. (20), area is just the number of image pixels $A(R)$ that make up the region R , N represents the value of area, which is equal to the number of elements in R ,

$$A(R) = |R| = N \quad (20)$$

The Gaussian region equation for polygons, as illustrated in Eq. (21), may be used to calculate the area of a connected region (without holes),

$$A(R) \approx \frac{1}{2} \left| \sum_{i=0}^{M-1} (u_i \cdot v_{i+1 \bmod M} - u_{i+1 \bmod M} \cdot v_i) \right| \quad (21)$$

where M is the vertices of the polygon, u_i and v_i are co-ordinates of the i -th vertex of the polygon.

3.9.6 Eccentricity

Eccentricity is a geometric property used in image analysis and classification to quantify the elongation or stretchiness of objects or regions within an image, which can be valuable in distinguishing between different classes or categories of objects based on their shapes. It is expressed as

$$\text{Ecc}(R) = \frac{a_1}{a_2} = \frac{\mu_{20} + \mu_{02} + \sqrt{(\mu_{20} - \mu_{02})^2 + 4\mu_{11}^2}}{\mu_{20} + \mu_{02} - \sqrt{(\mu_{20} - \mu_{02})^2 + 4\mu_{11}^2}} \quad (22)$$

where a_1 and a_2 are the major and minor axis lengths, respectively. μ_{20} , μ_{02} , μ_{11} are central moments. Feature extraction (GLCM) is a technique for suppressing the original image data set by measuring selected values or features that aid in classifying pictures. Color, form, and texture comprise the visual content of an image, but the semantic content is complicated and depends on the visual content^[32].

3.10 Classification

The input space is divided into numerous local regions by fuzzy clustering. The computation complexity is reduced based on Euclidean distance, and the fuzzy rule number is determined by a validity function based on both the separation and the compactness among clusters. This improved adaptive neuro-fuzzy neural network^[32] is used where all cell types from the microscopic blood cell dataset are examined. The dataset's microscopic blood images have a native resolution of 2592 pixel × 1944 pixel and are obtained with a Powershot G5 camera^[33].

(1) Architecture

To establish the architecture of an ANFIS, the first order Sugeno style Fuzzy Inference System (FIS) is initially introduced. The mapping from a given crisp input to a crisp output is controlled by fuzzy set theory in a first-order Sugeno style FIS model, as shown in the following:

If

$$x_1 \text{ is } A_1, x_2 \text{ is } A_2, \dots, x_n \text{ is } A_n \quad (23)$$

then

$$y = k_0 + k_1 k_1 + k_2 k_2 + \dots + k_n k_n \quad (24)$$

where $x_1, x_2, \dots, x_n$ represent input signals, $A_1, A_2, \dots, A_n$ represent fuzzy sets, and y represents the output variable^[31]. Find it in a fuzzy rule like this. The output variable^[32] is a first-order polynomial input variable.

(2) Description of the method

An ANFIS is an ANN that is functionally equivalent to a first-order Sugeno style FIS. An ANFIS model consists of one input layer, four hidden layers, and one output layer, for a total of six levels. Signal forwarding is the responsibility of every layer. A similar ANFIS model is shown in Fig. 6.

A network with two inputs, X and Y , and one output

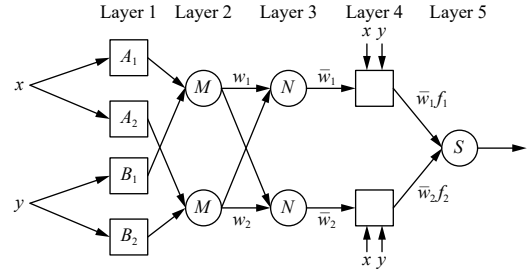


Fig. 6 Improved ANFIS architecture.

is assumed. The neurons in this layer are simply conveyed to the next layer from external input (crisp) signals, as illustrated in the following:

$$X_i^1 = Y_i^1 \quad (25)$$

In the input layer, X_i^1 is the input signal, and Y_i^1 is the output signal of neuron i . This layer's nodes adjust to a function parameter^[33]. Equations (26) and (27) show the output from each node as a degree of membership value that is supplied by the input of the membership functions. A Gaussian membership function, a modified bell membership function, or any form of membership function, for example, can be used to establish the architecture of an ANFIS, the first order Sugeno is as follows:

$$O_i^1 = \mu_{A_i}(x), \text{ for } i = 1, \text{ and } 2 \quad (26)$$

$$O_i^1 = \mu_{B_i}(y), \text{ for } i = 3, \text{ and } 4 \quad (27)$$

The inputs to node i are x and y , and the outputs are fuzzy membership grades of the inputs. Every node utilizes the Gaussian membership function indicated in Eq. (27) to compute the degree of membership of the input.

The fourth hidden layer is the defuzzification layer. A defuzzification neuron calculates the “weighted consequent value” of a given rule, and each node in this layer is an adaptive node to output with a node function denoted by

$$O_i^4 = \bar{w}_i f_i = \bar{w}_i (p_i x_1 + q_i x_2 + r_i x_3 + a_i x_4 + b_i x_5 + c_i x_6 + d_i x_7 + t_i) \quad (28)$$

where O_i^4 is the output of the i -th defuzzification neuron in the fourth hidden layer, $\bar{w}_i$ represents the normalized firing strength or output from the previous layer, which serves as the weight of the consequent parameters, f_i is the node function of the i -th neuron, calculated using the consequent parameters, and $p_i, q_i, r_i, a_i, b_i, c_i, d_i$, and t_i are the consequent parameters associated with the i -th neuron. Only one neuron in the

fourth layer is used to calculate the total of all defuzzification neurons' outputs, and the overall ANFIS output is created as a result^[33].

4 Experimental Result

The study's findings are given and analysed in this chapter with regard to the study's goal, which is to examine the impact of employing visual representations of indications in microscopic blood cells^[34–38]. The results and discussion related to the proposed work are also provided, including segmentation using enhances dual threshold, feature extraction using hybrid DWT classification, and I-ANFIS. This study examines several classification algorithms, such as relevance vector machines, adaptive neuro-fuzzy inference systems, and k-nearest neighbors, for the categorization of microscopic blood cell data.

Many computational analyses are carried out. The resulting prediction performances are comparable with those of already available methods. However, it is discovered that the suggested classification method, dubbed the I-ANFIS, performs significantly better than previous methods^[39].

4.1 Performance measure

The suggested work's findings help in analysing the process' efficiency using microscopic blood cell datasets. Performance measurement is the process of quantifying the efficiency and effectiveness of previous actions. To determine results, all performance measurement processes involve statistical modelling. A complete copy of an organization's performance can never be achieved since some parameters cannot be assessed directly and must be inferred by indirect observation, and a complete collection of records can never offer an evaluation without compression of crucial statistics.

Enhanced Dual Threshold Segmentation (EDTS) is

used for feature extraction to improve accuracy, precision, and recall. The results of comparing the F-measure to dual threshold segmentation accuracy are listed in Table 1. Feature extraction aims to make models simpler and clearer, improves classification performance, and provides clean and intelligible data. A comparison of classification accuracy using various feature extraction algorithms is shown in Table 1.

Table 2 shows the various segmentation and EDTS comparisons with Dual Threshold Segmentation (DTS) for all cell types, including band cells, metamyelocytes, myeloblasts, N. myelocytes, N. promyelocytes, and neutrophil cells.

Figure 7 shows the EDTS comparison of segmentation accuracy performance for all cell types.

Table 3 shows the comparison of the performance analysis of the proposed I-ANFIS.

The performance analysis presented by I-ANFIS is compared to several classification approaches, ANFIS, KNN, and RVM, shown Fig. 8.

The comparison of classification performance for various classification strategies is shown in Fig. 9. Table 4 compares the proposed I-ANFIS to ANFIS, KNN, and RVM in terms of performance.

4.2 Discussion

The proposed algorithm for classifications is tested on the microscopic blood cell datasets. The experiments have been carried out to assess the following criteria:

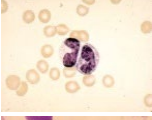
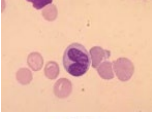
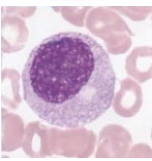
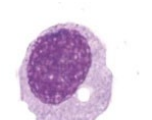
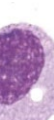
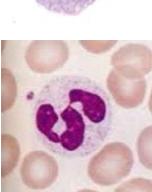
- To determine if the consistency measure and consistency factor are effective criteria for assessing the calibre of feature subsets produced.
- To evaluate the accuracy, sensitivity, specificity, and classification performance analysis of classifiers with respect to the features obtained from the segmentation.
- To evaluate the performance of the algorithm in terms of the classification accuracy needed to reach a good solution for leukemia detection,

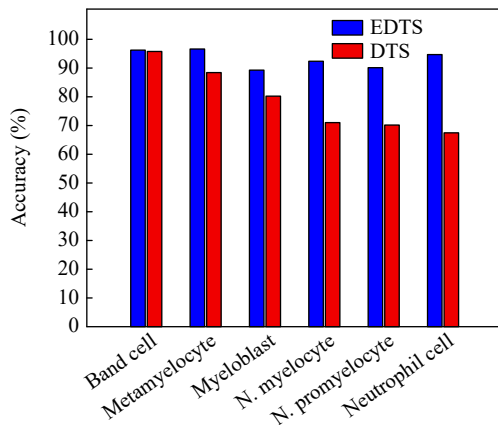
Table 1 Different segmentation and comparison of EDTS performance measure.

Cell type	DTS accuracy	EDTS accuracy	Precision	Recall	F-measure
Band cell	95.7264	96.3456	90.29	99.99	94.89
Metamyelocyte	88.0432	97.2896	98.55	99.45	99.00
Myeloblast	79.8480	89.3216	88.33	99.09	93.40
N. myelocyte	70.9344	92.5936	86.40	99.85	92.64
N. promyelocyte	70.0864	89.7856	87.60	99.96	93.37
Neutrophil cell	67.3392	94.8224	99.86	99.90	99.88

(%)

Table 2 Different segmentation and EDTS comparison for all cell types.

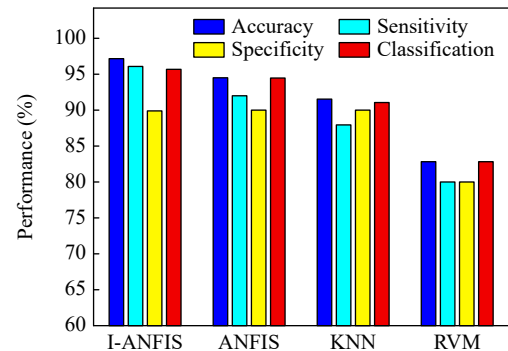
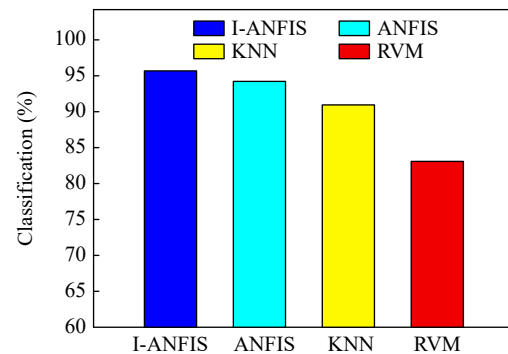
Cell type	Input image	DTS	EDTS
Band cell			
Metamyelocyte			
Myeloblast			
N.myelocyte			
N.promyelocyte			
Neutrophil cell			

**Fig. 7** Comparison of accuracy performance for EDTS.

Experiments demonstrate that applying a hybrid discrete wavelet transform for feature extraction in the pre-processing approach for improved dual threshold segmentation considerably improves classification accuracy. Also, any feature extraction strategy has significantly improved the classifier's accuracy compared to the accuracy without feature extraction. With the aim of testing if the feature extraction

Table 3 Comparison of performance analysis for the proposed I-ANFIS.

Algorithm	Accuracy	Sensitivity	Specificity
I-ANFIS	97.1429	0.9600	0.9000
ANFIS	94.2857	0.9200	0.9000
KNN	91.4286	0.8800	0.9000
RVM	82.8571	0.8000	0.8000

**Fig. 8** Comparison of performance analysis for the proposed technique.**Fig. 9** Comparison of classification performance for the proposed technique.**Table 4** Comparison of classification performance for the proposed I-ANFIS.

Algorithm	Accuracy
I-ANFIS	95.67
ANFIS	94.29
KNN	91.05
RVM	82.86

technique can provide the maximum accuracy for all cell types and microscopic datasets of the same domain, the best feature extraction strategy for a given dataset depends on the quantity, type, and occurrences of attributes. I-ANFIS can be used to increase the

proposed classifier's accuracy once the best feature extraction method for a specific dataset has been identified.

In the world of medical diagnostics, accuracy is crucial for diagnosing leukemia. Experiments demonstrate that using feature extraction as a pre-processing strategy improves classification accuracy significantly. It also concludes that using EDTS feature extraction improves the classifier's accuracy, sensitivity, specificity, and classification. This comparative analysis chapter discusses and displays the performance measure and performance bar.

Different classification algorithms are compared to the proposed I-ANFIS classification using microscopic blood cell datasets. The suggested classification approach, I-ANFIS, outperforms the competition in terms of accuracy, sensitivity, specificity, and classification metrics. This work aids in the diagnosis of leukemia by physicians and healthcare professionals.

5 Conclusion

In conclusion, this research presents a novel approach for discriminating between normal and acute lymphoblastic leukaemia in bone marrow samples. The proposed method incorporates an enhanced dual threshold segmentation technique, accurate nucleus and cytoplasm segmentation, feature extraction using hybrid discrete wavelet transform, and classification utilizing the I-ANFIS. By utilizing microscopic images of blood cells obtained from the ALL-IDB2 databases, the I-ANFIS machine learning system effectively classifies normal and leukemic blood cells based on the extracted characteristics. The proposed approach addresses the issue of class imbalance and demonstrates its efficacy through evaluation metrics such as accuracy, sensitivity, specificity, and classification performance comparison with existing methods. Looking ahead, future research can explore additional avenues and research directions within the scope of this I-ANFIS classification investigation. Specifically, adapting the proposed strategy to other machine learning approaches with additional adjustable parameters holds potential for further improvement and exploration. In summary, this research presents a comprehensive method for automatic segmentation, feature extraction, and classification of bone marrow samples to differentiate between normal and acute lymphoblastic leukaemia. The proposed approach offers enhanced classification performance, as well as

improved speed and accuracy, compared to alternative methods, making it a promising tool for medical diagnosis in this domain.

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